

# Exam Question Bank

Sampling, Circular Convolution, Z-Transforms  
and Linear Convolution

**Name:** \_\_\_\_\_

**Date:** \_\_\_\_\_

**Instructions.** Answer all ten questions. Show the necessary working and state units wherever applicable. Unless indices are explicitly given, the first entry of a sequence is at  $n = 0$ . Finite input sequences are zero outside their listed indices. For Question 2, use the Nyquist rate to obtain the sampling interval and sampled sequence.

## Part A — Questions only

1. State the **Nyquist sampling theorem**. Define **Nyquist rate** and **Nyquist interval**.

2. Given

$$x(t) = 4 + 3 \cos(200\pi t) + 2 \cos(1000\pi t),$$

determine the highest frequency component, Nyquist rate, sampling interval, and sampled sequence.

3. Given

$$x(t) = 5 \cos(60\pi t),$$

determine the Nyquist rate and obtain the sampled sequence at the threshold sampling frequency.

4. Find the **4-point circular convolution** of

$$x = \{1, 2, 3, 4\}, \quad h = \{2, 1\}.$$

5. Find the **3-point circular convolution** of

$$x = \{2, 1, 3\}, \quad h = \{1, 2\}.$$

**Part A — Questions only (continued)**

6. Find the Z-transform  $X(z)$  of

$$x = \{3, 2, 1\}_{n=-1,0,1}.$$

7. Find  $X(z)$  and  $H(z)$  for

$$x = \{1, 2, 3\}_{n=-1,0,1}, \quad h = \{2, 1\}_{n=0,1}.$$

Hence determine  $Y(z) = X(z)H(z)$ .

8. Find the **linear convolution**

$$y[n] = x[n] * h[n]$$

for

$$x = \{1, 2, 1\}, \quad h = \{2, 1\}.$$

9. Find the linear convolution of

$$x = \{2, 1, 3\}, \quad h = \{1, 2\}$$

using the **Z-transform method**.

10. Given

$$x(t) = 2 + 4 \cos(100\pi t) + 3 \cos(600\pi t) + 5 \cos(1200\pi t),$$

find the individual frequency components, maximum frequency, Nyquist rate, sampling interval, and sampled sequence at the Nyquist rate.





## Solution 1 — Nyquist sampling theorem

**Brief exam answer.** A continuous-time signal band-limited to a highest frequency  $f_m$  can be reconstructed exactly from its uniformly spaced samples if the sampling frequency is greater than twice  $f_m$ , with ideal reconstruction:

$$f_s > 2f_m.$$

The limiting rate is called the **Nyquist rate**. The corresponding limiting time between samples is the **Nyquist interval**:

$$f_N = 2f_m \text{ samples/s}, \quad T_N = \frac{1}{2f_m} \text{ s.}$$

**Exam convention used in these numericals.** The threshold is commonly written as  $f_s \geq 2f_m$ . When asked to sample “at the Nyquist rate” or “at the threshold”, set  $f_s = 2f_m$  and  $T_s = T_N$ . Equality is the limiting case, not a universal guarantee of reconstruction: a component  $\sin(2\pi f_m t)$  would give  $\sin(\pi n) = 0$  at every sample. Using  $f_s > 2f_m$  avoids this boundary issue.

### Meaning of the quantities

| Quantity      | Meaning                                                                        |
|---------------|--------------------------------------------------------------------------------|
| $f_m$         | Highest frequency actually present in the input, in Hz.                        |
| $f_s$         | Number of samples taken each second.                                           |
| $T_s = 1/f_s$ | Time between consecutive samples, in seconds.                                  |
| Band-limited  | No frequency components above the stated highest frequency.                    |
| Aliasing      | Different continuous-time frequencies become indistinguishable in the samples. |

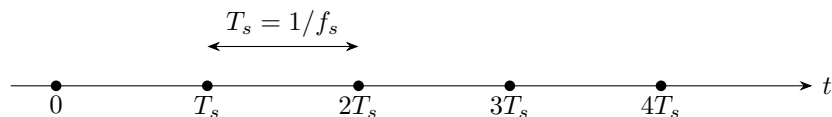
### How all three sampling numericals are solved

$$A \cos(\omega t) = A \cos(2\pi f t) \implies f = \frac{|\omega|}{2\pi}.$$

Find every frequency, select the largest, calculate  $2f_m$ , take its reciprocal, and replace **every**  $t$  by  $nT_s$ :

$$x[n] = x(nT_s), \quad n \in \mathbb{Z}.$$

A constant is a **DC component** at 0 Hz. Amplitude is the coefficient outside a cosine; it does not determine the frequency. The sampled sequence is indexed by  $n$ ; its first few entries do not mean it ends there.



## Solution 2 — Sampling a sum of two cosines

$$x(t) = 4 + 3 \cos(200\pi t) + 2 \cos(1000\pi t).$$

### 1. Convert angular frequencies to frequencies in Hz

Compare each cosine with  $A \cos(2\pi f t)$ , taking  $t$  in seconds.

| Term                | Amplitude | Angular frequency | Frequency                 |
|---------------------|-----------|-------------------|---------------------------|
| 4                   | 4         | 0                 | 0 Hz (DC)                 |
| $3 \cos(200\pi t)$  | 3         | $200\pi$ rad/s    | $200\pi/(2\pi) = 100$ Hz  |
| $2 \cos(1000\pi t)$ | 2         | $1000\pi$ rad/s   | $1000\pi/(2\pi) = 500$ Hz |

Thus  $f_m = 500$  Hz. The amplitudes 3 and 2 do not affect this choice.

### 2. Calculate the Nyquist rate and interval

$$f_N = 2f_m = 2(500) = 1000 \text{ samples/s.}$$

Using  $f_s = f_N$  as specified in the question-bank instructions,

$$T_s = T_N = \frac{1}{1000} \text{ s} = 0.001 \text{ s} = 1 \text{ ms.}$$

An actual sampling interval depends on the chosen  $f_s$ ; the value above is the interval *at the Nyquist rate*.

### 3. Replace every $t$ by $nT_s = n/1000$

$$\begin{aligned} x[n] &= 4 + 3 \cos\left(200\pi \frac{n}{1000}\right) + 2 \cos\left(1000\pi \frac{n}{1000}\right) \\ &= 4 + 3 \cos\left(\frac{\pi n}{5}\right) + 2 \cos(\pi n). \end{aligned}$$

Since  $\cos(\pi n) = (-1)^n$  for integer  $n$ ,

$$x[n] = 4 + 3 \cos\left(\frac{\pi n}{5}\right) + 2(-1)^n, \quad n \in \mathbb{Z}.$$

This formula is the exact sampled sequence. Decimal sample values are only approximations used for the graph.

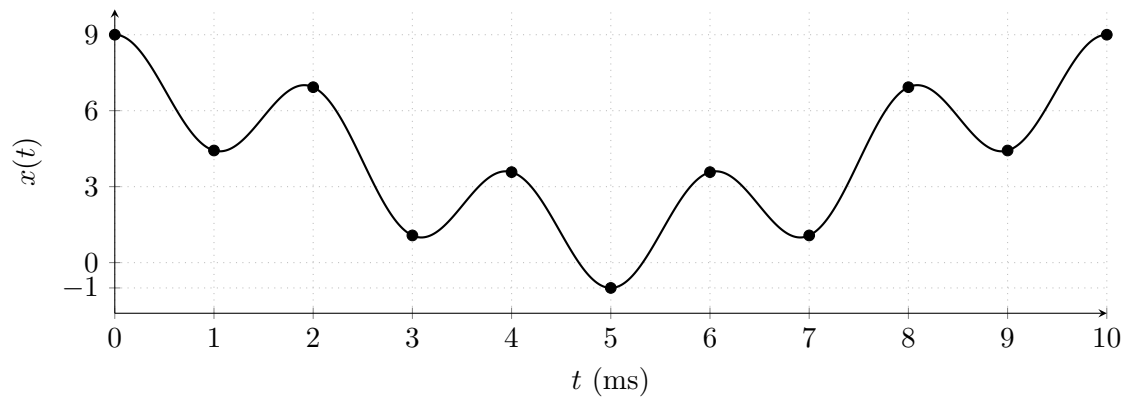
### 4. Check the first two samples

$$\begin{aligned} x[0] &= 4 + 3(1) + 2(1) = 9, \\ x[1] &= 4 + 3 \cos(\pi/5) - 2 \approx 4.427. \end{aligned}$$

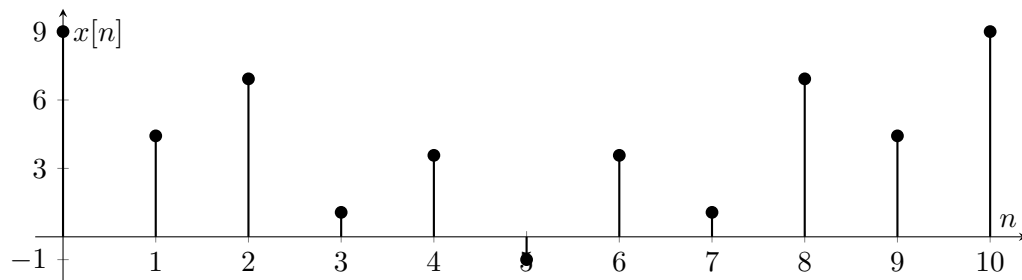
The discrete angular frequencies are  $\pi/5$  and  $\pi$  rad/sample. The sequence repeats every 10 samples, corresponding to  $10T_s = 10$  ms.

## Solution 2 — Graphs and sample-value check

**Continuous-time graph:** the line shows  $x(t)$  between sampling instants; the dots are its values at  $t = n$  ms. Only one 10 ms period is shown.



**Discrete-time graph:** stems show  $x[n]$ . The horizontal axis is an integer sample index; no connecting curve is implied.



| $n$             | 0     | 1     | 2     | 3     | 4     | 5  |
|-----------------|-------|-------|-------|-------|-------|----|
| $t = nT_s$ (ms) | 0     | 1     | 2     | 3     | 4     | 5  |
| $x[n]$          | 9     | 4.427 | 6.927 | 1.073 | 3.573 | -1 |
| $n$             | 6     | 7     | 8     | 9     | 10    |    |
| $t = nT_s$ (ms) | 6     | 7     | 8     | 9     | 10    |    |
| $x[n]$          | 3.573 | 1.073 | 6.927 | 4.427 | 9     |    |

**Reading the graph:**  $x[5] = -1$  is the value of the original signal at  $t = 5$  ms. Values at  $n = 0$  and  $n = 10$  agree because the sampled sequence repeats after ten samples.

### Solution 3 — A single cosine at the threshold

$$x(t) = 5 \cos(60\pi t).$$

#### 1. Identify the frequency and sampling rate

$$2\pi f = 60\pi \implies f = f_m = 30 \text{ Hz}, \quad \boxed{f_N = 2(30) = 60 \text{ samples/s.}}$$

At the threshold,  $f_s = 60$  samples/s and

$$T_s = \frac{1}{60} \text{ s} \approx 16.667 \text{ ms.}$$

#### 2. Sample by substitution

$$x[n] = 5 \cos\left(60\pi \frac{n}{60}\right) = 5 \cos(\pi n).$$

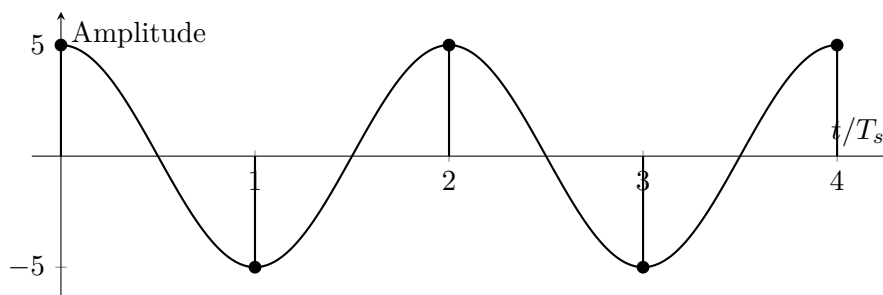
$$\boxed{x[n] = 5(-1)^n = \begin{cases} 5, & n \text{ even,} \\ -5, & n \text{ odd.} \end{cases}}$$

Starting at  $n = 0$ , the samples are  $\{5, -5, 5, -5, 5, -5, \dots\}$ . They continue for all integer  $n$  and have period 2 samples.

| $n$            | 0 | 1    | 2    | 3    | 4    |
|----------------|---|------|------|------|------|
| $t = nT_s$ (s) | 0 | 1/60 | 2/60 | 3/60 | 4/60 |
| $\cos(\pi n)$  | 1 | -1   | 1    | -1   | 1    |
| $x[n]$         | 5 | -5   | 5    | -5   | 5    |

#### 3. Graph: two samples per cosine cycle

The input period is  $1/30$  s, so  $T_s$  is half a period. The horizontal coordinate below is  $t/T_s$ ; integer values mark the sample instants.



The cosine in this question has zero phase, so its threshold samples alternate between its two extrema. The boundary qualification in Solution 1 still applies to general signals.



## Solution 4 — Four-point circular convolution

$$x = \{1, 2, 3, 4\}, \quad h = \{2, 1\}, \quad N = 4.$$

### 1. Write the indices and pad the shorter input

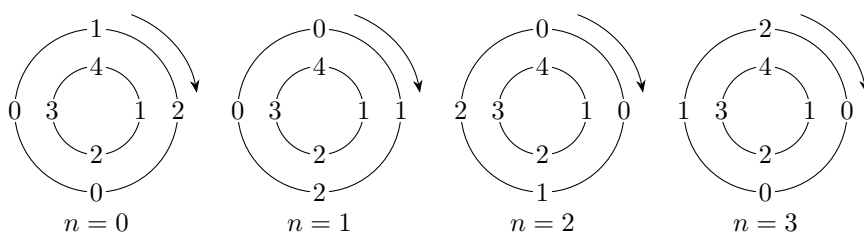
| $k$    | 0 | 1 | 2 | 3 |
|--------|---|---|---|---|
| $x[k]$ | 1 | 2 | 3 | 4 |
| $h[k]$ | 2 | 1 | 0 | 0 |

Append zeros **after** the listed entries:  $h = \{2, 1, 0, 0\}$ . The zeros belong to indices 2 and 3. Drawing the circle in a different direction does not move them to the beginning or change  $h[0]$ .

### 2. Set up the circle-and-rotate method

$$y_4[n] = \sum_{k=0}^3 x[k] h[(n-k) \bmod 4], \quad n = 0, 1, 2, 3.$$

Place  $x$  clockwise on the inner ring, starting at the right: 1, 2, 3, 4. Initially place  $h$  anticlockwise on the outer ring, starting at the right: 2, 1, 0, 0. Read aligned products and add. Rotate only the outer ring one position clockwise for each new output.



**Reading direction:** right, down, left, up corresponds to fixed  $k = 0, 1, 2, 3$ . The inner ring stays fixed in every diagram. The outer values are  $h[(n-k) \bmod 4]$ , not simply  $h[k]$ .

### 3. Multiply the aligned values and add

| $n$ | Outer values in fixed $k$ order | Calculation                      |
|-----|---------------------------------|----------------------------------|
| 0   | (2, 0, 0, 1)                    | $1(2) + 2(0) + 3(0) + 4(1) = 6$  |
| 1   | (1, 2, 0, 0)                    | $1(1) + 2(2) + 3(0) + 4(0) = 5$  |
| 2   | (0, 1, 2, 0)                    | $1(0) + 2(1) + 3(2) + 4(0) = 8$  |
| 3   | (0, 0, 1, 2)                    | $1(0) + 2(0) + 3(1) + 4(2) = 11$ |

$$y_4 = \{6, 5, 8, 11\}_{n=0,1,2,3}.$$

For example, at  $n = 0$ ,  $h[-1 \bmod 4] = h[3] = 0$  and  $h[-3 \bmod 4] = h[1] = 1$ . Modulo indexing wraps around the circle; it does not discard negative indices.

## Solution 4 — Z-transform verification and graph

### 4. Multiply the Z-transforms to find the linear result first

$$X(z) = 1 + 2z^{-1} + 3z^{-2} + 4z^{-3}, \quad H(z) = 2 + z^{-1}.$$

$$\begin{aligned} X(z)H(z) &= \underbrace{2 + 4z^{-1} + 6z^{-2} + 8z^{-3}}_{2X(z)} \\ &\quad + \underbrace{z^{-1} + 2z^{-2} + 3z^{-3} + 4z^{-4}}_{z^{-1}X(z)} \\ &= 2 + 5z^{-1} + 8z^{-2} + 11z^{-3} + 4z^{-4}. \end{aligned}$$

Ordinary multiplication gives the **linear** convolution:

$$y_{\text{lin}} = \{2, 5, 8, 11, 4\}_{n=0,1,2,3,4}.$$

### 5. Wrap the linear result into four positions

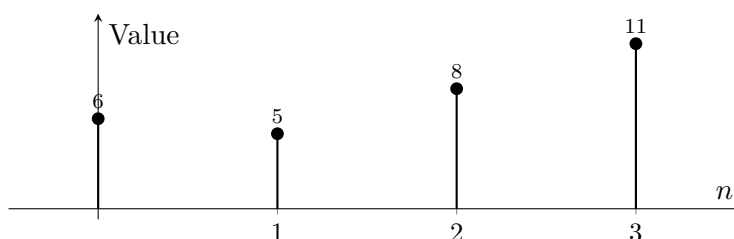
The requested answer has exactly  $N = 4$  entries. Fold each coefficient into the index with the same remainder modulo 4. In this example, index 4 wraps to index 0:

| Circular index | Linear entries to add                           | Value |
|----------------|-------------------------------------------------|-------|
| 0              | $y_{\text{lin}}[0] + y_{\text{lin}}[4] = 2 + 4$ | 6     |
| 1              | $y_{\text{lin}}[1]$                             | 5     |
| 2              | $y_{\text{lin}}[2]$                             | 8     |
| 3              | $y_{\text{lin}}[3]$                             | 11    |

$$y_4 = \{6, 5, 8, 11\}.$$

This agrees with the circle method. Stopping at the five-entry polynomial result would answer a *linear* convolution question instead.

One block of the four-point circular output



### Sum check:

$$6 + 5 + 8 + 11 = 30 = (1 + 2 + 3 + 4)(2 + 1).$$

The four-point output is one block. If interpreted as a circular sequence, it repeats with period 4. To obtain the full linear convolution by a circular-convolution calculation, the padding length would have to be at least  $4 + 2 - 1 = 5$ , instead of the 4 requested here.

## Solution 5 — Three-point circular convolution

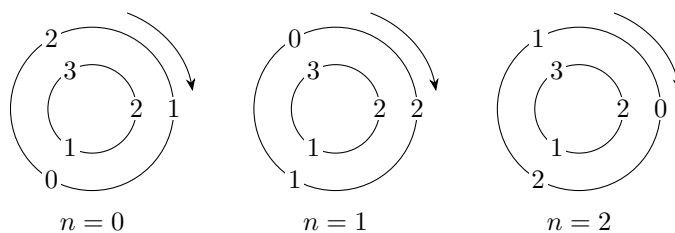
$$x = \{2, 1, 3\}, \quad h = \{1, 2\}, \quad N = 3.$$

### 1. Pad, reverse circularly, and rotate

Append one zero:  $h = \{1, 2, 0\}$  at indices 0, 1, 2. Use

$$y_3[n] = \sum_{k=0}^2 x[k]h[(n-k) \bmod 3].$$

Put  $x$  clockwise on the inner ring and padded  $h$  anticlockwise on the outer ring, both starting at the right. Read right, lower left, upper left for  $k = 0, 1, 2$ . Rotate the outer ring clockwise by one position per output.



| $n$ | Outer values | Aligned products and sum |
|-----|--------------|--------------------------|
| 0   | (1, 0, 2)    | $2(1) + 1(0) + 3(2) = 8$ |
| 1   | (2, 1, 0)    | $2(2) + 1(1) + 3(0) = 5$ |
| 2   | (0, 2, 1)    | $2(0) + 1(2) + 3(1) = 5$ |

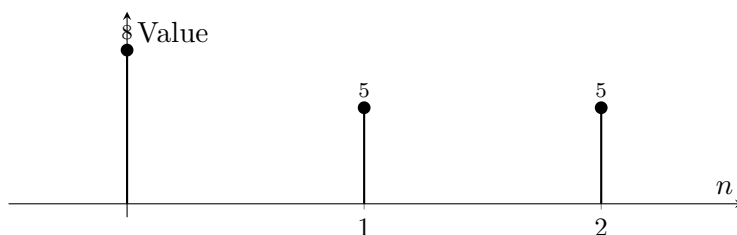
$$y_3 = \{8, 5, 5\}_{n=0,1,2}.$$

### 2. Verify by multiplication and wrapping

$$\begin{aligned} X(z)H(z) &= (2 + z^{-1} + 3z^{-2})(1 + 2z^{-1}) \\ &= 2 + 5z^{-1} + 5z^{-2} + 6z^{-3}. \end{aligned}$$

The linear result is  $\{2, 5, 5, 6\}$ . For  $N = 3$ , index 3 wraps to 0, so the circular result is  $\{2 + 6, 5, 5\} = \{8, 5, 5\}$ .

One block of the three-point circular output



**Sum check:**  $8 + 5 + 5 = 18 = (2 + 1 + 3)(1 + 2)$ . Compare Solution 9: the same inputs give a different answer when the question asks for *linear* convolution.

## Solution 6 — Z-transform with a negative starting index

$$x = \{3, 2, 1\}_{n=-1,0,1}.$$

### 1. Associate each entry with its stated index

$$x[-1] = 3, \quad x[0] = 2, \quad x[1] = 1.$$

The first entry is at  $n = -1$ , not at  $n = 0$ . All other samples are zero.

### 2. Apply the definition term by term

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n}.$$

For this finite sequence, only the three listed terms survive:

| Index $n$ | Value $x[n]$ | Power $z^{-n}$  | Term $x[n]z^{-n}$ |
|-----------|--------------|-----------------|-------------------|
| -1        | 3            | $z^{-(-1)} = z$ | $3z$              |
| 0         | 2            | $z^0 = 1$       | 2                 |
| 1         | 1            | $z^{-1}$        | $z^{-1}$          |

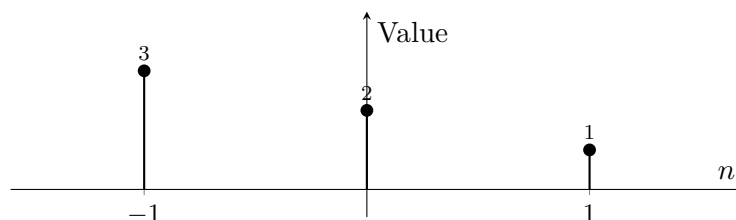
$$\begin{aligned} X(z) &= x[-1]z^{-(-1)} + x[0]z^0 + x[1]z^{-1} \\ &= 3z + 2 + z^{-1}. \end{aligned}$$

$$X(z) = 3z + 2 + z^{-1}.$$

### 3. Read the expression back to check the index signs

| Term               | Sample index | Sample value |
|--------------------|--------------|--------------|
| $3z = 3z^{-(-1)}$  | -1           | 3            |
| $2 = 2z^0$         | 0            | 2            |
| $z^{-1} = 1z^{-1}$ | 1            | 1            |

The input sequence: the middle entry is at the origin



**Key rule:** the exponent is the *negative* of the sample index. A positive power of  $z$  represents a negative-time sample. Writing  $3 + 2z^{-1} + z^{-2}$  would shift this sequence one sample to the right and would not represent the given indices.

## Solution 7 — Multiply two Z-transforms

$$x = \{1, 2, 3\}_{n=-1,0,1}, \quad h = \{2, 1\}_{n=0,1}.$$

1. Write each sample with its power of  $z$

$$\begin{aligned} X(z) &= 1z^{-(-1)} + 2z^0 + 3z^{-1} = \boxed{z + 2 + 3z^{-1}}, \\ H(z) &= 2z^0 + 1z^{-1} = \boxed{2 + z^{-1}}. \end{aligned}$$

2. Multiply every term, adding exponents

$$\begin{aligned} Y(z) &= X(z)H(z) = (z + 2 + 3z^{-1})(2 + z^{-1}) \\ &= z(2) + z(z^{-1}) + 2(2) + 2(z^{-1}) + 3z^{-1}(2) + 3z^{-1}(z^{-1}) \\ &= 2z + 1 + 4 + 2z^{-1} + 6z^{-1} + 3z^{-2}. \end{aligned}$$

Here  $zz^{-1} = z^0 = 1$  and  $z^{-1}z^{-1} = z^{-2}$ .

3. Collect only terms with the same power

$$Y(z) = 2z + (1 + 4) + (2 + 6)z^{-1} + 3z^{-2}.$$

$$\boxed{Y(z) = 2z + 5 + 8z^{-1} + 3z^{-2}}.$$

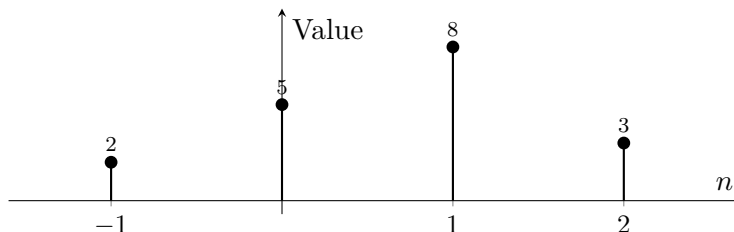
4. Read back the coefficients as an indexing check

Although the question stops at  $Y(z)$ , reading it back verifies the multiplication:

| Power              | $z^1$ | $z^0$ | $z^{-1}$ | $z^{-2}$ |
|--------------------|-------|-------|----------|----------|
| Index $n$          | -1    | 0     | 1        | 2        |
| Coefficient $y[n]$ | 2     | 5     | 8        | 3        |

Thus the associated linear convolution is  $y = \{2, 5, 8, 3\}_{n=-1,0,1,2}$ . Its first index is  $-1 + 0 = -1$  and its last is  $1 + 1 = 2$ , giving  $3 + 2 - 1 = 4$  entries.

Coefficient sequence represented by  $Y(z)$



**Check at  $z = 1$ :**  $Y(1) = 2 + 5 + 8 + 3 = 18 = X(1)H(1) = 6(3)$ .

## Solution 8 — Linear convolution

$$x = \{1, 2, 1\}_{n=0,1,2}, \quad h = \{2, 1\}_{n=0,1}.$$

### 1. Determine the output indices

The output length is  $L + M - 1 = 3 + 2 - 1 = 4$ , with indices  $0 + 0 = 0$  through  $2 + 1 = 3$ . Inputs outside their listed indices are zero.

### 2. Use the Z-transform multiplication method

$$X(z) = 1 + 2z^{-1} + z^{-2}, \quad H(z) = 2 + z^{-1}.$$

$$\begin{aligned} Y(z) &= (1 + 2z^{-1} + z^{-2})(2 + z^{-1}) \\ &= (2 + 4z^{-1} + 2z^{-2}) + (z^{-1} + 2z^{-2} + z^{-3}) \\ &= 2 + 5z^{-1} + 4z^{-2} + z^{-3}. \end{aligned}$$

Read the coefficient of  $z^{-n}$  as  $y[n]$ :

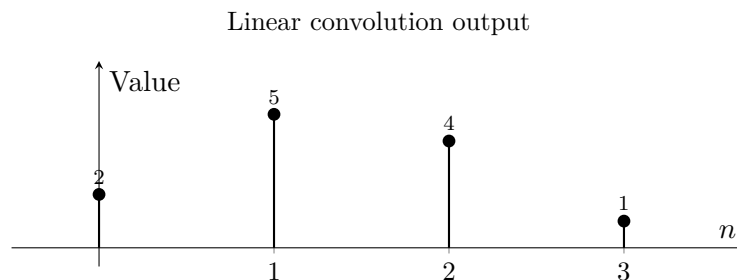
$$y = \{2, 5, 4, 1\}_{n=0,1,2,3}.$$

### 3. Verify each entry directly

Linear convolution uses  $y[n] = \sum_k x[k]h[n-k]$ . Only overlapping nonzero samples contribute:

$$\begin{aligned} y[0] &= x[0]h[0] = 1(2) = 2, \\ y[1] &= x[0]h[1] + x[1]h[0] = 1(1) + 2(2) = 5, \\ y[2] &= x[1]h[1] + x[2]h[0] = 2(1) + 1(2) = 4, \\ y[3] &= x[2]h[1] = 1(1) = 1. \end{aligned}$$

All other  $y[n]$  are zero. There is no modulo operation and no wrap-around.



**Sum check:**  $2 + 5 + 4 + 1 = 12 = (1 + 2 + 1)(2 + 1)$ .

**Why multiplication works:** for discrete-time linear convolution, the convolution property is  $\mathcal{Z}\{x*h\} = X(z)H(z)$ . It is the same method even when the question does not explicitly request a Z-transform approach.

## Solution 9 — Linear convolution using Z-transforms

$$x = \{2, 1, 3\}_{n=0,1,2}, \quad h = \{1, 2\}_{n=0,1}.$$

### 1. Write the transforms

$$X(z) = 2 + z^{-1} + 3z^{-2}, \quad H(z) = 1 + 2z^{-1}.$$

### 2. Multiply every term and collect equal powers

$$\begin{aligned} Y(z) &= (2 + z^{-1} + 3z^{-2})(1 + 2z^{-1}) \\ &= 2 + 4z^{-1} + z^{-1} + 2z^{-2} + 3z^{-2} + 6z^{-3} \\ &= \boxed{2 + 5z^{-1} + 5z^{-2} + 6z^{-3}}. \end{aligned}$$

### 3. Read the coefficient at each index

| Index $n$ | 0      | 1         | 2         | 3         |
|-----------|--------|-----------|-----------|-----------|
| Term      | $2z^0$ | $5z^{-1}$ | $5z^{-2}$ | $6z^{-3}$ |
| $y[n]$    | 2      | 5         | 5         | 6         |

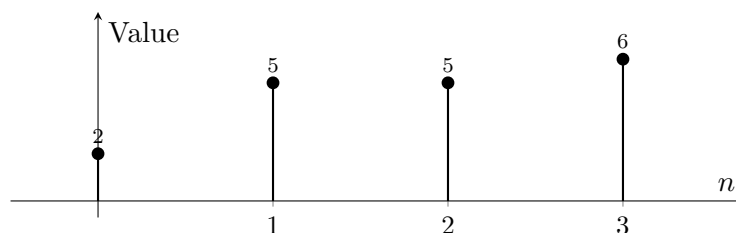
$$\boxed{y = \{2, 5, 5, 6\}_{n=0,1,2,3}}.$$

There are  $3 + 2 - 1 = 4$  entries, and  $y[n] = 0$  outside  $0 \leq n \leq 3$ .

### 4. Check the result by ordinary convolution

$$\begin{aligned} y[0] &= 2(1) = 2, & y[1] &= 2(2) + 1(1) = 5, \\ y[2] &= 1(2) + 3(1) = 5, & y[3] &= 3(2) = 6. \end{aligned}$$

Linear output: retain the entry at  $n = 3$



**Sum check:**  $2 + 5 + 5 + 6 = 18 = (2 + 1 + 3)(1 + 2)$ .

| Same inputs | Required operation           | Answer           |
|-------------|------------------------------|------------------|
| Question 5  | 3-point circular convolution | $\{8, 5, 5\}$    |
| Question 9  | Linear convolution           | $\{2, 5, 5, 6\}$ |

For the circular result, 6 at index 3 folds into index 0. For the linear result, it stays at index 3. Thus ordinary Z-transform multiplication needs an additional wrapping step only when a circular result is requested.

## Solution 10 — Sampling a three-frequency signal

$$x(t) = 2 + 4 \cos(100\pi t) + 3 \cos(600\pi t) + 5 \cos(1200\pi t).$$

### 1. Identify every frequency component

Use  $f = \omega/(2\pi)$  for each cosine; the constant is DC.

| Term                | Amplitude | $\omega$ (rad/s) | $f$ (Hz)               |
|---------------------|-----------|------------------|------------------------|
| 2                   | 2         | 0                | 0 (DC)                 |
| $4 \cos(100\pi t)$  | 4         | $100\pi$         | $100\pi/(2\pi) = 50$   |
| $3 \cos(600\pi t)$  | 3         | $600\pi$         | $600\pi/(2\pi) = 300$  |
| $5 \cos(1200\pi t)$ | 5         | $1200\pi$        | $1200\pi/(2\pi) = 600$ |

$$f \in \{0, 50, 300, 600\} \text{ Hz}, \quad f_m = 600 \text{ Hz}.$$

The highest frequency is 600 Hz, not  $1200\pi$  Hz:  $1200\pi$  is an angular frequency in rad/s.

### 2. Find the Nyquist rate and interval

$$f_N = 2f_m = 1200 \text{ samples/s}.$$

At the Nyquist rate,  $f_s = 1200$  samples/s, so

$$T_s = T_N = \frac{1}{1200} \text{ s} = \frac{5}{6} \text{ ms} \approx 0.833333 \text{ ms}.$$

### 3. Replace every $t$ by $n/1200$

$$\begin{aligned} x[n] &= 2 + 4 \cos\left(100\pi \frac{n}{1200}\right) + 3 \cos\left(600\pi \frac{n}{1200}\right) + 5 \cos\left(1200\pi \frac{n}{1200}\right) \\ &= 2 + 4 \cos\left(\frac{\pi n}{12}\right) + 3 \cos\left(\frac{\pi n}{2}\right) + 5 \cos(\pi n). \end{aligned}$$

$$x[n] = 2 + 4 \cos\left(\frac{\pi n}{12}\right) + 3 \cos\left(\frac{\pi n}{2}\right) + 5(-1)^n, \quad n \in \mathbb{Z}.$$

The three discrete angular frequencies are  $\pi/12$ ,  $\pi/2$ , and  $\pi$  rad/sample. The DC term remains 2.

### 4. Establish the repetition period

The three cosine sequences have periods 24, 4, and 2 samples, respectively. The whole sequence therefore has period 24 samples, corresponding to  $24T_s = 20$  ms. A portion of this sequence is tabulated and plotted on the next page.



## Solution 10 — Sample values and graphs

Two explicit checks, using the exact formula:

$$\begin{aligned} x[0] &= 2 + 4 + 3 + 5 = 14, \\ x[2] &= 2 + 4 \cos(\pi/6) + 3 \cos(\pi) + 5 \cos(2\pi) \\ &= 2 + 2\sqrt{3} - 3 + 5 = 4 + 2\sqrt{3} \approx 7.464. \end{aligned}$$

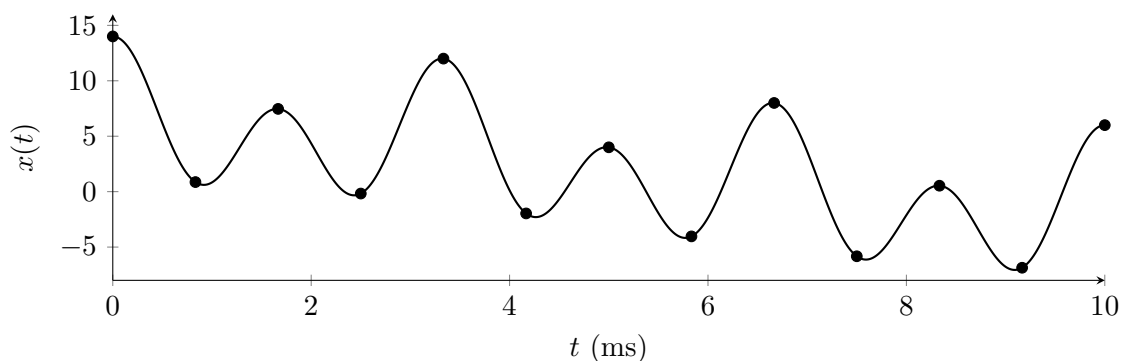
| $n$      | 0  | 1     | 2     | 3      | 4     | 5      | 6     |
|----------|----|-------|-------|--------|-------|--------|-------|
| $t$ (ms) | 0  | 0.833 | 1.667 | 2.500  | 3.333 | 4.167  | 5.000 |
| $x[n]$   | 14 | 0.864 | 7.464 | -0.172 | 12    | -1.965 | 4     |

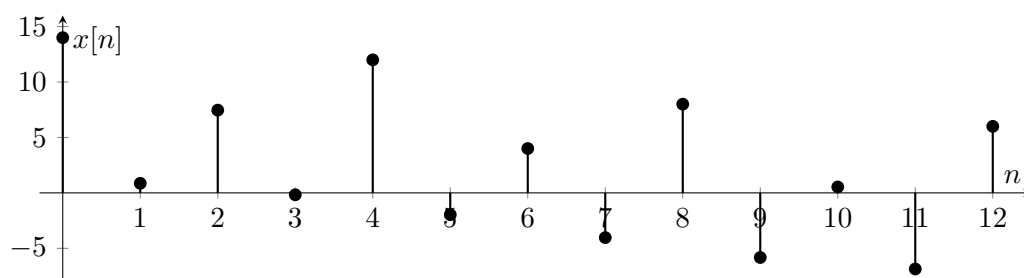
| $n$      | 7      | 8     | 9      | 10    | 11     | 12     |
|----------|--------|-------|--------|-------|--------|--------|
| $t$ (ms) | 5.833  | 6.667 | 7.500  | 8.333 | 9.167  | 10.000 |
| $x[n]$   | -4.035 | 8     | -5.828 | 0.536 | -6.864 | 6      |

Values above are rounded to three decimal places; the exact formula is the answer.

Continuous-time signal with sample points



Discrete-time sequence: samples 0 through 12



Both graphs show the same sample values: the time coordinate of sample  $n$  is  $t = n/1200$  seconds. Only half of a 24-sample period is displayed; the sequence continues.